Exercise 18A

Solution 1:

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

(i) The first eight natural numbers are 1, 2, 3, 4, 5, 6, 7 and 8.

Mean of these numbers:

$$= \frac{1+2+3+4+5+6+7+8}{8}$$

$$= \frac{36}{8}$$

$$= 4.5$$

(ii) The first ten odd numbers are 1, 3, 5, 7, 9, 11, 13, 15, 17 and 19.

Mean of these numbers:

$$= \frac{1+3+5+7+9+11+13+15+17+19}{10}$$

$$= \frac{100}{10}$$

$$= 10$$

(iii) The first seven multiples of 5 are 5, 10, 15, 20, 25, 30 and 35.

Mean of these numbers:

$$= \frac{(5+10+15+20+25+30+35)}{7}$$

$$= \frac{140}{7}$$

$$= 20$$

(iv) The factors of 20 are 1, 2, 4, 5, 10 and 20.

Mean of these numbers:

$= \frac{1+2+4+5+10+20}{6}$

$= \frac{42}{6}$

$= 7$

(v) The prime numbers between 50 and 80 are 53, 59, 61, 67, 71, 73 and 79.

Mean of these numbers:

$= \frac{53+59+61+67+71+73+79}{7}$

$= \frac{463}{7}$

$= 66.14$

Solution 2: Numbers of children in 10 families = 2, 4, 3, 4, 2, 0, 3, 5, 1 and 6.

Thus, we have:

$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$

$= \frac{2+4+3+4+2+0+3+5+1+6}{10}$

$= \frac{30}{10}$

$= 3$

Solution 3: Numbers of books issued in the school library: 105, 216, 322, 167, 273, 405 and 346

Thus, we have:

$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$

$= \frac{105+216+322+167+273+405+346}{7}$

$= \frac{1834}{7}$

$= 262$

Solution 4: Daily minimum temperatures = 35.5, 30.8, 27.3, 32.1, 23.8 and 29.9

Thus, we have:

$$\text{Mean temperature} = \frac{35.5 + 30.8 + 27.3 + 32.1 + 23.8 + 29.9}{6} = \frac{179.4}{6} = 29.9°\text{F}$$

Solution 5: We know that,

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

The first five observations are x, $x + 2$, $x + 4$, $x + 6$ and $x + 8$.

$$\text{Mean of these numbers} = \frac{x + (x+2) + (x+4) + (x+6) + (x+8)}{5}$$

$$\Rightarrow 13 = \frac{5x + 20}{5}$$

$$\Rightarrow 13 \times 5 = 5x + 20$$

$$\Rightarrow 65 = 5x + 20$$

$$\Rightarrow 5x = 65 - 20$$

$$\Rightarrow 5x = 45$$

$$\Rightarrow x = \frac{45}{5}$$

$$\Rightarrow x = 9$$

Hence, the value of x is 9.

Now, the last three observations are 13, 15 and 17.

Mean of these observations

$$= \frac{13 + 15 + 17}{3}$$

$$= \frac{45}{3}$$

$$= 15$$

Hence, the mean of the last three observations is 15.

Solution 6: The individual weights of five boys are 51 kg, 45 kg, 49 kg, 46 kg and 44 kg.

Now,

Let the weight of the sixth boy be x kg.

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Also,

Given mean = 48 kg

Thus, we have:

$$48 = \frac{51 + 45 + 49 + 46 + 44 + x}{6}$$

$$\Rightarrow 288 = 235 + x$$

$$\Rightarrow x = 53$$

Therefore, the sixth boy weighs 53 kg.

Solution 7: Let the marks scored by 50 students be $x_1, x_2, \ldots x_{50}$.

Mean = 39

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$39 = \frac{x_1 + x_2 + \cdots + x_{50}}{50}$$

$$\Rightarrow x_1 + x_2 + \cdots + x_{50} = 1950 \qquad \ldots \text{(i)}$$

Also, a score of 43 was misread as 23.

$$\therefore \text{New Mean} = \frac{(x_1 + x_2 + \cdots + x_{50}) - 23 + 43}{50}$$

$$= \frac{1950 - 23 + 43}{50} \qquad [\text{ using (i)}]$$

$$= \frac{1970}{50}$$

$$= 39.4$$

Solution 8: Let the numbers be $x_1, x_2, \ldots x_{24}$.

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$35 = \frac{x_1 + x_2 + \cdots + x_{24}}{24}$$

$$\Rightarrow 840 = x_1 + x_2 + \cdots + x_{24} \qquad \ldots \text{(i)}$$

After addition, the new numbers become $(x_1+3), (x_2+3), \ldots (x_{24}+3)$.

New mean:

$$= \frac{(x_1 + 3)(x_2 + 3) + \cdots (x_{24} + 3)}{24}$$

$$= \frac{(x_1 + x_2 + \cdots + x_{24}) + (24 \times 3)}{24}$$

$$= \frac{840 + 72}{24} \qquad \text{[using (i)]}$$

$$= \frac{912}{24}$$

$$= 38$$

Solution 9: Let the numbers be $x_1, x_2, \ldots x_{20}$.

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$43 = \frac{x_1 + x_2 + \cdots + x_{20}}{20}$$

$$\Rightarrow 860 = x_1 + x_2 + \cdots + x_{20} \qquad \ldots \text{(i)}$$

Numbers after subtraction: $(x_1-6), (x_2-6), \ldots (x_{20}-6)$

$$\therefore \text{New Mean} = \frac{(x_1 - 6) + (x_2 - 6) + \cdots + (x_{20} - 6)}{20}$$

$$= \frac{(x_1 + x_2 + \cdots + x_{20}) - (20 \times 6)}{20}$$

$= \frac{860 - 120}{20}$ [using (i)]

$= 37$

Solution 10: Let the numbers be $x_1, x_2, \ldots x_{15}$

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$27 = \frac{x_1 + x_2 + \cdots + x_{15}}{15}$$

$\Rightarrow x_1 + x_2 + \cdots + x_{15} = 405$

After multiplication, the numbers become $4x_1, 4x_2, \ldots 4x_{15}$

$$\therefore \text{New Mean} = \frac{4x_1 + 4x_2 + \cdots + 4x_{15}}{15}$$

$$= \frac{4(x_1 + x_2 + \cdots + x_{15})}{15}$$

$= \frac{4 \times 405}{15}$ [using (i)]

$$= \frac{1620}{15}$$

$= 108$

Solution 11: Let the numbers be $x_1, x_2, \ldots x_{12}$.

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$40 = \frac{x_1 + x_2 + \cdots + x_{12}}{12}$$

$\Rightarrow x_1 + x_2 + \cdots + x_{12} = 480$ ... (i)

After division, the numbers become:

$$\frac{x_1}{8}, \frac{x_2}{8}, \ldots \ldots \ldots, \frac{x_{12}}{8}$$

$\therefore$ New mean $= \dfrac{\left(\frac{x_1}{8}\right) + \left(\frac{x_2}{8}\right) + \cdots \left(\frac{x_{12}}{8}\right)}{12}$

$= \dfrac{x_1 + x_2 + \cdots + x_{12}}{12 \times 8}$

$= \dfrac{480}{96}$ [using(i)]

$= 5$

Solution 12: Let the numbers be $x_1, x_2, \ldots x_{20}$.

We know:

Mean $= \dfrac{\text{Sum of observations}}{\text{Number of observations}}$

Thus, we have:

$18 = \dfrac{x_1 + x_2 + \cdots + x_{20}}{20}$

$\Rightarrow x_1 + x_2 + \cdots + x_{20} = 360$ ... (i)

New numbers are:

$(x_1 + 3), (x_2 + 3), \ldots (x_{10} + 3), x_{11}, \ldots x_{20}$

$\therefore$ New Mean:

$= \dfrac{(x_1 + 3) + (x_2 + 3) + \cdots + (x_{10} + 3) + x_{11} + \cdots + x_{20}}{20}$

$= \dfrac{(x_1 + x_2 + \cdots + x_{10}) + (3 \times 10) + x_{11} + \cdots x_{20}}{20}$

$= \dfrac{(x_1 + x_2 + \cdots + x_{20}) + 30}{20}$

$= \dfrac{360 + 30}{20}$ [using (i)]

$= \dfrac{390}{20}$

$= 19.5$

Solution 13: Let the numbers be $x_1, x_2, \ldots, x_6$.

Mean $= 23$

We know:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Thus, we have:

$$23 = \frac{x_1 + x_2 + \cdots + x_6}{6}$$

$\Rightarrow x_1 + x_2 + \cdots + x_6 = 138$ ... (i)

If one number, say, x_6, is excluded, then we have:

$$20 = \frac{x_1 + x_2 + \cdots + x_5}{5}$$

$\Rightarrow x_1 + x_2 + \cdots + x_5 = 100$ ... (ii)

Using (i) and (ii), we get:

$138 = x_1 + x_2 + \cdots + x_5 + x_6$

$\Rightarrow 138 = 100 + x_6$ [using (ii)]

$\Rightarrow x_6 = 38$

Thus, the excluded number is 38

Or

Mean of 6 numbers = 23

Sum of 6 numbers = $(23 \times 6) = 138$

Again, mean of 5 numbers =20

Sum of 5 numbers = $(20 \times 5) = 100$

∴ The excluded number = (sum of 6 numbers) - (sum of 5 numbers)

= (138 - 100)

= 38

∴ The excluded number = 38.

Solution 14: Mean height of 30 boys = 150 cm

⇒ Total height of 30 boys = 150×30

= 4500 cm

Correct sum = 4500 - incorrect value + correct value

= 4500 - 135 + 165

= 4530

$\therefore$ Corrected Mean $= \dfrac{\text{Correct sum}}{30} = \dfrac{4530}{30} = 151$ cm

Solution 15: Mean weight of 34 students = 46.5 kg

Sum of the weights of 34 students = (46.5×34) kg = 1581

Increase in the mean weight when the weight of the teacher is included = 500 g = 0.5 kg

$\therefore$ New mean weight = $(46.5 + 0.5)$ kg = 47 kg

Now,

Let the weight of the teacher be x kg.

Thus, we have:

$$\frac{\text{Sum of the weights of 34 students} + \text{Weight of the teacher}}{35} = 47$$

$$\Rightarrow \frac{1581 + x}{35} = 47$$

$$\Rightarrow 1581 + x = 1645$$

$$\Rightarrow x = 64$$

Therefore, the weight of the teacher is 64 kg.

Solution 16: Mean weight of 36 students = 41 kg

Sum of the weights of 36 students = (41×36) kg = 1476

Decrease in the mean when one of the students left the class = 200 g = 0.2 kg

Mean weight of 35 students = $(41 - 0.2)$ kg = 40.8 kg

Now,

Let the weight of the student who left the class be x kg.

$$\frac{\text{Thus, we have: Sum of the weights of 36 students} - x}{35} = 40.8$$

$$\Rightarrow \frac{1476 - x}{35} = 40.8$$

$$\Rightarrow 1476 - x = 1428$$

$$\Rightarrow x = 48$$

Hence, the weight of the student who left the class is 48 kg.

Solution 17: Average weight of 39 students = 40 kg

Sum of the weights of 39 students = (40×39) kg = 1560

Decrease in the average when new student is admitted in the class = 200 g = 0.2 kg

$\therefore$ New average weight = $(40 - 0.2)$ kg = 39.8 kg

Now,

Let the weight of the new student be x kg.

Thus, we have:

$$\frac{\text{Sum of the weights of 39 students} + x}{40} = 39.8$$

$$\Rightarrow \frac{1560 + x}{40} = 39.8$$

$$\Rightarrow 1560 + x = 1592$$

$$\Rightarrow x = 32$$

Therefore, the weight of the new student is 32 kg.

Solution 18: The increase in the average of 10 oarsmen = 1.5 kg

Total weight increased = (1.5×10) kg=15 kg

Since the man weighing 58 kg has been replaced,

Weight of the new man = (58+15) kg = 73kg.

Solution 19: Mean of 8 numbers = 35

Total sum of 8 numbers = $35 \times 8 = 280$

Since One number is excluded, New mean = $35 - 3 = 32$

$\therefore$ Total sum of 7 numbers = $32 \times 7 = 224$

$\therefore$ the excluded number = Sum of 8 numbers − Sum of 7 numbers

$= 280 - 224$

$= 56$

Therefore, the excluded number is 56.

Solution 20:

Mean of 150 items = 60

Total Sum of 150 items = 150x60 = 9000

Correct sum of items = [(sum of 150 items) − (sum of wrong items) + (sum of right items)]

= [9000 − (52 + 8) + (152 + 88)]

= [9000 − 60 + 240]

= 9180

Therefore, the correct mean is 61.2.

Solution 21: Mean of 31 results = 60

Sum of 31 results = 31 × 60 = 1860

Mean of the first 16 results = 58

Sum of the first 16 results = 58 × 16 = 928

Mean of the last 16 results = 62

Sum of the last 16 results = 62 × 16 = 992

Value of the 16th result = (Sum of the first 16 results + Sum of the last 16 results) −- Sum of 31 results

= (928 + 992) −- 1860

= 1920 − 1860

= 60

Solution 22: Mean of 11 numbers = 42

Sum of 11 numbers = 42 × 11 = 462

Mean of the first 6 numbers = 37

Sum of the first 6 numbers = 37 × 6 = 222

Mean of the last 6 numbers = 46

Sum of the last 6 numbers = 46 × 6 = 276

∴ 6th number = [(Sum of the first 6 numbers + Sum of the last 6 numbers) − Sum of 11 numbers]

$= [(222 + 276) - 462]$

$= [498 - 462]$

$= 36$

Hence, the 6th number is 36.

Solution 23: Mean weight of 25 students = 52 kg

Sum of the weights of 25 students = (52×25) kg = 1300 kg

Mean weight of the first 13 students = 48 kg

Sum of the weights of the first 13 students = (48×13) kg = 624 kg

Mean weight of the last 13 students = 55 kg

Sum of the weights of the last 13 students = (55×13) kg =715 kg

Weight of the 13th student = (Sum of the weights of the first 13 students + Sum of the weights of the last 13 students) − Sum of the weights of 25 students

$= [(624+715) - 1300]$ kg

$= 39$ kg

Therefore, the weight of the 13th student is 39 kg.

Solution 24: Mean score of 25 observations = 80

Total score of 25 observations = $80 \times 25 = 2000$

Mean score of 55 observations = 60

Total score of 55 observations = $60 \times 55 = 3300$

Total no. of observations = 25+55 =80 observations

$\therefore$ Total score = $2000 + 3300 = 5300$

$\therefore$ Mean score $= \frac{5300}{80} = 66.25$

Therefore, the mean score of the whole set of observations is 66.25.

Solution 25:

Marks scored by Arun in English = 36

Marks scored by Arun in Hindi = 44

Marks scored by Arun in mathematics = 75

Marks scored by Arun in science = x

Average marks = 50

Thus, we have:

$$\text{Average marks} = \frac{36 + 44 + 75 + x}{4}$$

$$\Rightarrow 50 = \frac{155 + x}{4}$$

$$\Rightarrow 200 = 155 + x$$

$$\Rightarrow x = 200 - 155$$

$$\Rightarrow x = 45$$

∴ Marks scored by Arun in science = 45

Solution 26:

Let the distance from the starting point to the island be x km.

Speed of the ship sailing out to the island = 15 km/h

Speed of the ship sailing back to the starting point = 10 km/h

We know:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Time taken by the ship to travel from the starting point to the island $= \frac{x}{15}$ h

Time taken by the ship to travel from the island to the starting point $= \frac{x}{10}$ h

$$\text{Total time taken} = \frac{x}{15} + \frac{x}{10} = \frac{2x + 3x}{30} = \frac{5x}{30} = \frac{x}{6}$$

Total distance covered $= x + x = 2x$ km.

$$\therefore \text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{2x}{\frac{x}{6}}$$

$= \dfrac{2x \times 6}{x}$

$= 12\ km/h$

Therefore, the average speed of the ship in the whole journey was 12 km/h.

Solution 27: Total number of students = 50

Total number of girls = 50 − 40 = 10

Average weight of the class = 44 kg

Total weight of 50 students= 44 × 50 kg = 2200kg

Average weight of 10 girls = 40 kg

Total weight of 10 girls = 40 ×10 kg = 400 kg

∴ Total weight of 40 boys = 2200 − 400 kg = 1800 kg

∴ the average weight of the boys $= \left(\dfrac{1800}{40}\right)$ kg = 45 kg

Solution 28:

The aggregate yearly expenditure of a family = (18720 × 3) + (20340 × 4) + (21708 × 5)

= (56160 + 81360 + 108540)

= ₹ 246060

The total savings during the year = ₹ 35340

The average yearly income = ₹ 246060 + 35340

= ₹ 281400

∴ The average monthly income of the family $= \dfrac{281400}{12}$ = ₹ 23450

Hence, The average monthly income of the family is ₹ 23450.

Solution 29: Total salary of 75 workers = ₹ 5680 × 75

= ₹ 426000

Total salary of 25 workers = ₹ 5400×25

= ₹ 135000

Total salary of 30 workers = ₹ 5700×30

= ₹ 171000

No. of remaining workers = $75 - (25 + 30)$

= 20

Total salary of 20 workers = ₹ $(426000 - 135000 - 171000)$

= ₹ 120000

$\therefore$ The mean weekly payment of the 20 workers $= \dfrac{120000}{20}$

= ₹ 6000

Hence, the mean weekly payment of the remaining workers is ₹ 6000.

Solution 30: Let the ratio of number of boys to the number of girls be x : 1.

Then,

Sum of marks of boys = 70x

Sum of marks of girls = $73 \times 1 = 73$

And,

sum of marks of boys and girls = $71 \times (x + 1)$

$\Rightarrow 70x + 73 = 71(x + 1)$

$\Rightarrow 70x + 73 = 71x + 71$

$\Rightarrow x = 2$

Hence, the ratio of number of boys to the number of girls is 2 : 1.

Solution 31: Mean monthly salary of 20 workers = ₹ 45900

$\Rightarrow$ Total monthly salary of 20 workers = ₹ (20×45900) = ₹ 918000

Mean monthly salary of 20 workers + manager = ₹ 49200

$\Rightarrow$ Total monthly salary of 20 workers + manager = ₹ (21×49200) = ₹ 1033200

Therefore, manager's monthly salary = ₹ (1033200 - 918000) = ₹ 115200

Exercise 18B

Solution 1:

Variable (x_i)	Frequency (f_i)	$f_i \times x_i$
4	4	16
6	8	48
8	14	112
10	11	110
12	3	36
	$\sum f_i = 40$	$\sum (f_i x_i) = 322$

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{322}{40} = 8.05$$

Solution 2: We will make the following table:

Weight (x_i)	No. of Workers (f_i)	$(f_i x_i)$
60	4	240
63	3	189
66	2	132
69	2	138
72	1	72
	$\sum f_i = 12$	$\sum f_i x_i = 771$

Thus, we have:

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{771}{12} = 64.250\ kg$$

Solution 3:

Diameter (in mm) (x_i)	Number of screws (f_i)	$f_i \times x_i$
34	5	170

37	10	370
40	17	680
43	12	516
46	6	276
	$\sum f_i = 50$	$\sum (f_i x_i) = 2012$

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{2012}{50} = 40.24\ mm$$

Hence, the mean diameter of the heads of the screws is 40.24 mm.

Solution 4: We will make the following table:

Age (x_i)	Frequency (f_i)	($f_i x_i$)
15	3	45
16	8	128
17	9	153
18	11	198
19	6	114
20	3	60
	$\sum f_i = 40$	$\sum (f_i x_i) = 698$

Thus, we have:

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{698}{40} = 17.45$$

Solution 5: We will make the following table:

Variable (x_i)	Frequency (f_i)	($f_i x_i$)
10	7	70

30	8	240
50	10	500
70	15	1050
89	10	890
	$\sum f_i = 50$	$\sum (f_i x_i) = 2750$

Thus, we have:

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{2750}{50} = 55$$

Solution 6: We will make the following table:

Daily wages (in ₹) (x_i)	Number of workers (f_i)	($f_i x_i$)
250	8	2000
300	11	3300
350	6	2100
400	10	4000
450	5	2250
	$\sum f_i = 40$	$\sum (f_i x_i) = 13650$

Thus, we have:

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum f_i} = \frac{13650}{40} = ₹\ 341.25$$

Solution 7:

Variable (x_i)	Frequency (f_i)	($f_i x_i$)
10	6	60
15	8	120
20	p	20p
25	10	250

30	6	180
	$\sum f_i = 30 + p$	$\sum (f_i x_i) = 610 + 20p$

Thus, we have:

Mean $= 20.2$

$\Rightarrow \dfrac{\sum(f_i x_i)}{\sum f_i} = 20.2$

$\Rightarrow \dfrac{610 + 20p}{30 + p} = 20.2$

$\Rightarrow 610 + 20p = 20.2(30 + p)$

$\Rightarrow 610 + 20p = 606 + 20.2p$

$\Rightarrow 20.2p - 20p = 610 - 606$

$\Rightarrow 0.2p = 4$

$\Rightarrow p = \dfrac{4}{0.2}$

$\Rightarrow p = \dfrac{4 \times 10}{2}$

$\Rightarrow p = 20$

Solution 8: We will make the following table:

(x_i)	(f_i)	$(f_i x_i)$
3	6	18
5	8	40
7	15	105
9	p	$9p$
11	8	88
13	4	52
	$\sum f_i = 41 + p$	$\sum (f_i x_i) = 303 + 9p$

Thus, we have:

Mean = 8

$$\Rightarrow \frac{\sum(f_i x_i)}{\sum f_i} = 8$$

$$\Rightarrow \frac{303 + 9p}{41 + p} = 8$$

$\Rightarrow 303 + 9p = 8(41 + p)$

$\Rightarrow 303 + 9p = 328 + 8p$

$\Rightarrow 9p - 8p = 328 - 303$

$\Rightarrow p = 25$

Solution 9: We will prepare the following table:

(x_i)	(f_i)	$(f_i x_i)$
15	8	120
20	7	140
25	p	$25p$
30	14	420
35	15	525
40	6	240
	$\sum f_i = 50 + p$	$\sum(f_i x_i) = 1445 + 25p$

Thus, we have:

Mean = 28.25

$$\Rightarrow \frac{\sum(f_i x_i)}{\sum f_i} = 28.25$$

$$\Rightarrow \frac{1445 + 25p}{50 + p} = 28.25$$

$\Rightarrow 1445 + 25p = 28.25(50 + p)$

$\Rightarrow 1445 + 25p = 1412.50 + 28.25p$

$\Rightarrow 28.25p - 25p = 1445 - 1412.50$

$\Rightarrow 3.25p = 32.5$

$\Rightarrow p = \frac{32.5}{3.25}$

$\Rightarrow p = 10$

Solution 10: We will make the following table:

(x_i)	(f_i)	$(f_i x_i)$
8	12	96
12	16	192
15	20	300
p	24	$24p$
20	16	320
25	8	200
30	4	120
	$\sum f_i = 100$	$\sum (f_i x_i) = 1228 + 24p$

Thus, we have:

Mean $= 16.6$

$\Rightarrow \frac{\sum (f_i x_i)}{\sum f_i} = 16.6$

$\Rightarrow \frac{1228 + 24p}{100} = 16.6$

$\Rightarrow 1228 + 24p = 1660$

$\Rightarrow 24p = 1660 - 1228$

$\Rightarrow 24p = 432$

$\Rightarrow p = \frac{432}{24}$

$\Rightarrow p = 18$

Solution 11:

x_i	f_i	$f_i x_i$
10	4	40
20	f_1	$20f_1$
30	8	240
40	f_2	$40f_2$
50	3	150
60	4	240
	$35 = \sum f_i = 19 + f_1 + f_2$	$\sum f_i x_i = 670 + 20f_1 + 40f_2$

$\sum f_i = 19 + f_1 + f_2$

$\Rightarrow 35 = 19 + f_1 + f_2$

$\Rightarrow f_1 + f_2 = 16$ ... (i)

Mean $= 34$

$\Rightarrow \frac{\sum f_i x_i}{\sum f_i} = 34$

$\Rightarrow \frac{670 + 20f_1 + 40f_2}{35} = 34$

$\Rightarrow \frac{335 + 10f_1 + 20f_2}{35} = 17$

$\Rightarrow 335 + 10f_1 + 20f_2 = 595$

$\Rightarrow 10f_1 + 20f_2 = 260$

$\Rightarrow f_1 + 2f_2 = 26$ ... (ii)

Subtracting (i) from (ii), we get

$f_2 = 10$

Putting the value of f_2 in (1), we get

$f_1 + f_2 = 16$

$\Rightarrow f_1 + 10 = 16$

$\Rightarrow f_1 = 6$

Hence, the value of f_1 and f_2 is 6 and 10, respectively.

Solution 12: We will prepare the following table:

(x_i)	(f_i)	$(f_i x_i)$
10	17	170
30	f_1	$30f_1$
50	32	1600
70	f_2	$70f_2$
90	19	1710
	$\sum f_i = 120$	$\sum (f_i x_i) = 3480 + 30f_1 + 70f_2$

Thus, we have:

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\Rightarrow 50 = \frac{3480 + 30f_1 + 70f_2}{120}$$

$\Rightarrow 600 = 3480 + 30f_1 + 70f_2$

$\Rightarrow 30f_1 + 70f_2 = 2520$ ... (i)

Also,

Given:

$17 + f_1 + 32 + f_2 + 19 = 120$

$\Rightarrow 68 + f_1 + f_2 = 120$

$\Rightarrow f_1 + f_2 = 52$

or, $f_2 = 52 - f_1$ ...(ii)

By putting the value of f_2 in (i), we get:

$2520 = 30f_1 + 70(52 - f_1)$

$\Rightarrow 2520 = 30f_1 + 3640 - 70f_1$

$\Rightarrow 40f_1 = 1120$

$\Rightarrow f_1 = 28$

Substituting the value in (ii), we get:

$f_2 = 52 - f_1$

$= 52 - 28$

$= 24$

Solution 13:

(x_i)	(f_i)	(f_ix_i)
15	2	30
17	3	51
19	4	76
20 + p	5p	$100p + 5p^2$
23	6	138
	$\sum f_i = 15 + 5p$	$\sum (f_ix_i) = 5p^2 + 100p + 295$

Thus, we have:

Mean = 20

$$\Rightarrow \frac{\sum(f_ix_i)}{\sum f_i} = 20$$

$$\Rightarrow \frac{5p^2 + 100p + 295}{15 + 5p} = 20$$

$\Rightarrow 5p^2 + 100p + 295 = 300 + 100p$

$\Rightarrow 5p^2 = 5$

$\Rightarrow p^2 = \pm 1$

Solution 14:

(x_i)	(f_i)	(f_ix_i)

10	17	170
30	5a + 3	150a + 90
50	32	1600
70	7a −11	$490a - 770$
90	19	1710
	$\sum f_i = 60 + 12a$	$\sum(f_i x_i) = 2800 + 640a$

Thus, we have:

Mean = 50

$\Rightarrow \frac{\sum(f_i x_i)}{\sum f_i} = 50$

$\Rightarrow \frac{2800 + 640a}{60 + 12a} = 50$

$\Rightarrow 2800 + 640a = 3000 + 600a$

$\Rightarrow 40a = 200$

$\Rightarrow a = 5$

Hence,

Frequency corresponding to $30 = 5a + 3 = 5 \times 5 + 3 = 28$

Frequency corresponding to $70 = 7a - 11 = 7 \times 5 - 11 = 24$

Exercise 18C

Solution 1:

(i) Arranging the numbers in ascending order, we get:

2, 2, 3, 5, 7, 9, 9, 10, 11

Here, n is 9, which is an odd number.

If n is an odd number, we have:

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Now,

$$\text{Median} = \text{Value of } \left(\frac{9+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of } \left(\frac{10}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of the } 5^{\text{th}} \text{ observation}$$

$$= 7$$

(ii) Arranging the numbers in ascending order, we get:

6, 8, 9, 15, 16, 18, 21, 22, 25

Here, n is 9, which is an odd number.

If n is an odd number, we have:

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Now,

$$\text{Median} = \text{Value of } \left(\frac{9+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of } \left(\frac{10}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of the } 5^{\text{th}} \text{ observation}$$

$$= 16$$

(iii) Arranging the numbers in ascending order, we get:

6, 8, 9, 13, 15, 16, 18, 20, 21, 22, 25

Here, n is 11, which is an odd number.

If n is an odd number, we have:

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Now,

$$\text{Median} = \text{Value of } \left(\frac{11+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of } \left(\frac{12}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of the } 6^{\text{th}} \text{ observation}$$

$$= 16$$

(iv) Arranging the numbers in ascending order, we get:

0, 1, 2, 2, 3, 4, 4, 5, 5, 7, 8, 9, 10

Here, n is 13, which is an odd number.

If n is an odd number, we have:

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Now,

$$\text{Median} = \text{Value of } \left(\frac{13+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of } \left(\frac{14}{2}\right) \text{th observation}$$

$$= \text{Value of the } 7^{\text{th}} \text{ observation}$$

$$= 4$$

Solution 2:

(i) Arranging the numbers in ascending order, we get:

9, 10, 17, 19, 21, 22, 32, 35

Here, n is 8, which is an even number.

If n is an even number, we have:

Median $= \text{Mean of } \left(\frac{n}{2}\right) \text{th} \;\&\; \left(\frac{n}{2}+1\right)^{\text{th}}$ observations

Now,

Median $= \text{Mean of } \left(\frac{8}{2}\right)^{\text{th}} \;\&\; \left(\frac{8}{2}+1\right)^{\text{th}}$ observations

$= \text{Mean of } \left(\frac{8}{2}\right)^{\text{th}} \;\&\; \left(\frac{10}{2}\right)^{\text{th}}$ observations

$=$ Mean of the 4^{th} & 5^{th} observations

$= \frac{1}{2}(19 + 21)$

$= 20$

(ii) Arranging the numbers in ascending order, we get:

29, 35, 51, 55, 60, 63, 72, 82, 85, 91

Here, n is 10, which is an even number.

If n is an even number, we have:

Median $= \text{Mean of } \left(\frac{n}{2}\right) \text{th} \;\&\; \left(\frac{n}{2}+1\right)^{\text{th}}$ observations

Now,

Median $= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \;\&\; \left(\frac{10}{2}+1\right)^{\text{th}}$ observations

$= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \;\&\; \left(\frac{12}{2}\right)^{\text{th}}$ observations

$=$ Mean of the 5^{th} & 6^{th} observations

$= \frac{1}{2}(60 + 63)$

$= 61.5$

(iii) Arranging the numbers in ascending order, we get:

3, 4, 9, 10, 12, 15, 17, 27, 47, 48, 75, 81

Here, n is 12, which is an even number.

If n is an even number, we have:

$$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)\text{th } \& \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observations}$$

Now,

$$\text{Median} = \text{Mean of } \left(\frac{12}{2}\right)^{\text{th}} \ \& \left(\frac{12}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{12}{2}\right)^{\text{th}} \ \& \left(\frac{14}{2}\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of the } 6^{\text{th}} \ \& \ 7^{\text{th}} \text{ observations}$$

$$= \frac{1}{2}(15+17)$$

$$= 16$$

Solution 3: Arranging the marks of 15 students in ascending order, we get:

17, 17, 19, 19, 20, 21, 22, 23, 24, 25, 26, 29, 31, 35, 40

Here, n is 15, which is an odd number.

We know:

$$\text{Median} = \text{ Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Now,

$$\text{Median} = \text{Value of } \left(\frac{15+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of } \left(\frac{16}{2}\right)^{\text{th}} \text{ observation}$$

$$= \text{Value of the } 8^{\text{th}} \text{ observation}$$

$$= 23$$

Solution 4: Arranging the given data in ascending order:

144, 145, 147, 148, 149, 150, 152, 155, 160

Number of terms = 9 (odd)

$$\therefore \text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$$

$$= \left(\frac{9+1}{2}\right)^{\text{th}} \text{ term}$$

$= \left(\frac{10}{2}\right)^{\text{th}}$ term

$= 5^{\text{th}}$ term

$= 149$

Hence, the median height is 149.

Solution 5: Arranging the weights (in kg) in ascending order, we have:

9.8, 10.6, 12.7, 13.4, 14.3, 15, 16.5, 17.2

Here, n is 8, which is an even number.

Thus, we have:

$$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)^{\text{th}} \ \& \ \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{8}{2}\right)^{\text{th}} \ \& \ \left(\frac{8}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{8}{2}\right)^{\text{th}} \ \& \ \left(\frac{10}{2}\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of the } 4^{\text{th}} \ \& \ 5^{\text{th}} \text{ observations}$$

$$= \frac{1}{2}(13.4 + 14.3)$$

$$= 13.85$$

Hence, the median weight is 13.85 kg.

Solution 6: Arranging the ages (in years) in ascending order, we have:

32, 34, 36, 37, 40, 44, 47, 50, 53, 54

Here, n is 10, which is an even number.

Thus, we have:

$$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)^{\text{th}} \ \& \ \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \ \& \ \left(\frac{10}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \ \& \ \left(\frac{12}{2}\right)^{\text{th}} \text{ observations}$$

$= \text{Mean of the } 5^{\text{th}} \text{ \& } 6^{\text{th}} \text{ observations}$

$$= \frac{1}{2}(40 + 44)$$

$= 42$

Hence, the median age is 42 years.

Solution 7: 10, 13, 15, 18, x+1, x+3, 30, 32, 35 and 41 are arranged in ascending order.

Median = 24

We have to find the value of x.

Here, n is 10, which is an even number.

Thus, we have:

$$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)^{\text{th}} \text{ \& } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \text{ \& } \left(\frac{10}{2} + 1\right)^{\text{th}} \text{ observations}$$

$$= \text{Mean of } \left(\frac{10}{2}\right)^{\text{th}} \text{ \& } \left(\frac{12}{2}\right)^{\text{th}} \text{ observations}$$

$= \text{Mean of the } 5^{\text{th}} \text{ \& } 6^{\text{th}} \text{ observations}$

$$= \frac{1}{2}(2x + 4)$$

$= (x + 2)$

Given:

Median = 24

$\Rightarrow x + 2 = 24$

$\Rightarrow x = 22$

Solution 8: Arranging the given data in ascending order:

26, 29, 42, 53, x, x+ 2, 70, 75, 82, 93

Number of terms = 10 (even)

$$\therefore \text{ Median} = \text{mean of } \left[\left(\frac{n}{2}\right)^{\text{th}} \text{ term and } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}\right]$$

$\Rightarrow 65 = \text{mean of}\left[\left(\frac{10}{2}\right)^{\text{th}} \text{ term and } \left(\frac{10}{2}+1\right)^{\text{th}} \text{ term}\right]$

$\Rightarrow 65 = \text{mean of}\left[\left(\frac{10}{2}\right)^{\text{th}} \text{ term and } \left(\frac{12}{2}\right)^{\text{th}} \text{ term}\right]$

$\Rightarrow 65 = \text{mean of}\left[(5)^{\text{th}} \text{ term and } (6)^{\text{th}} \text{ term}\right]$

$\Rightarrow 65 = \text{mean of}\,[x \text{ and } x+2]$

$\Rightarrow 65 = \frac{x+x+2}{2}$

$\Rightarrow 65 \times 2 = 2x + 2$

$\Rightarrow 130 = 2x + 2$

$\Rightarrow 2x = 130 - 2$

$\Rightarrow 2x = 128$

$\Rightarrow x = 64$

Hence, the value of x is 64.

Solution 9: Arranging the given data in ascending order:

8, 11, 12, $(2x - 8)$, $(2x + 10)$, 35, 42, 50

Number of terms = 8 (even)

$\therefore$ Median = mean of $\left[\left(\frac{n}{2}\right)^{\text{th}} \text{ term and } \left(\frac{n}{2}+1\right)^{\text{th}} \text{ term}\right]$

$\Rightarrow 25 = \text{mean of}\left[\left(\frac{8}{2}\right)^{\text{th}} \text{ term and } \left(\frac{8}{2}+1\right)^{\text{th}} \text{ term}\right]$

$\Rightarrow 25 = \text{mean of}\left[\left(\frac{8}{2}\right)^{\text{th}} \text{ term and } \left(\frac{10}{2}\right)^{\text{th}} \text{ term}\right]$

$\Rightarrow 25 = \text{mean of}\left[(4)^{\text{th}} \text{ term and } (5)^{\text{th}} \text{ term}\right]$

$\Rightarrow 25 = \text{mean of}\,[2x - 8 \text{ and } 2x + 10]$

$\Rightarrow 25 = \frac{2x - 8 + 2x + 10}{2}$

$\Rightarrow 25 \times 2 = 4x + 2$

$\Rightarrow 50 = 4x + 2$

$\Rightarrow 4x = 50 - 2$

$\Rightarrow 4x = 48$

$\Rightarrow x = 12$

Hence, the value of x is 12.

Solution 10: Arranging the given data in ascending order:

33, 35, 41, 46, 55, 58, 64, 77, 87, 90, 92

Number of terms = 11 (odd)

$\therefore$ Median $= \left(\frac{n+1}{2}\right)^{\text{th}}$ term

$= \left(\frac{11+1}{2}\right)^{\text{th}}$ term

$= \left(\frac{12}{2}\right)^{\text{th}}$ term

$= 6^{\text{th}}$ term

$= 58$

Hence, the median of the data is 58.

Now, In the above data, if 41 and 55 are replaced by 61 and 75 respectively.

Then, new data in ascending order is:

33, 35, 46, 58, 61, 64, 75, 77, 87, 90, 92

Number of terms = 11 (odd)

$\therefore$ Median $= \left(\frac{n+1}{2}\right)^{\text{th}}$ term

$= \left(\frac{11+1}{2}\right)^{\text{th}}$ term

$= \left(\frac{12}{2}\right)^{\text{th}}$ term

$= 6^{\text{th}}$ term

$= 64$

Hence, the new median of the data is 64.

Exercise 18D

Solution 1: On arranging the items in ascending order, we get:

0, 0, 1, 2, 3, 4, 5, 5, 6, 6, 6, 6

Clearly, 6 occurs maximum number of times.

$\therefore$ Mode = 6

Solution 2: On arranging the values in ascending order, we get:

15, 20, 22, 23, 25, 25, 25, 27, 40

Clearly, 25 occurs maximum number of times.

$\therefore$ Mode = 25

Solution 3: On arranging the shoe sizes in ascending order, we get:

1, 1, 2, 3, 3, 4, 5, 5, 6, 6, 7, 8, 9, 9, 9, 9, 9

Clearly, 9 occurs maximum number of times.

$\therefore$ Mode = 9

Solution 4: On arranging the runs in ascending order, we get:

9, 19, 27, 28, 30, 32, 35, 50, 50, 50, 50, 60

Clearly, 50 occurs maximum number of times.

$\therefore$ Modal score = 50

Solution 5: We know that,

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

The given data is 3, 21, 25, 17, (x + 3), 19, (x – 4).

$$\text{Mean of the given data} = \frac{3 + 21 + 25 + 17 + (x + 3) + 19 + (x - 4)}{7}$$

$$\Rightarrow 18 = \frac{84 + 2x}{7}$$

$$\Rightarrow 18(7) = 84 + 2x$$

$\Rightarrow 126 - 84 = 2x$

$\Rightarrow 2x = 42$

$\Rightarrow x = 21$

Hence, the value of x is 21.

Now, the given data is 3, 21, 25, 17, 24, 19, 17

Arranging this data in ascending order:

3, 17, 17, 19, 21, 24, 25

Here, 17 occurs maximum number of times.

$\therefore$ Mode = 17

Hence, the mode of the data is 17.

Solution 6: Arranging the given data in ascending order:

52, 53, 54, 54, (2x + 1), 55, 55, 56, 57

Number of terms = 9 (odd)

$\therefore$ Median $= \left(\frac{n+1}{2}\right)^{th}$ term

$\Rightarrow 55 = \left(\frac{9+1}{2}\right)^{th}$ term

$\Rightarrow 55 = (5)^{th}$ term

$\Rightarrow 55 = 2x + 1$

$\Rightarrow 2x = 55 - 1$

$\Rightarrow 2x = 54$

$\Rightarrow x = 27$

Hence, the value of x is 27.

Arranging the given data in ascending order:

52, 53, 54, 54, 55, 55, 55, 56, 57

Here, 55 occurs maximum number of times.

$\therefore$ Mode = 55

Hence, the mode of the data is 55.

Solution 7: Given: Mode = 25

$\therefore$ 25 occurs maximum number of times.

Arranging the given data in ascending order:

15, 20, 22, 23, 24, $x + 3$, 25, 26, 27, 40

$\therefore x + 3 = 25$

$\Rightarrow x = 25 - 3$

$\Rightarrow x = 22$

Hence, the value of x is 22.

Arranging the given data in ascending order:

15, 20, 22, 23, 24, 25, 25, 26, 27, 40

Number of terms = 10 (even)

$\therefore$ Median $= \text{mean of } \left[\left(\frac{n}{2}\right)^{th} \text{ term and } \left(\frac{n}{2}+1\right)^{th} \text{ term}\right]$

$= \text{mean of } \left[\left(\frac{10}{2}\right)^{th} \text{ term and } \left(\frac{10}{2}+1\right)^{th} \text{ term}\right]$

$= \text{mean of } \left[(5)^{th} \text{ term and } \left(\frac{10+2}{2}\right)^{th} \text{ term}\right]$

$= \text{mean of } \left[(5)^{th} \text{ term and } \left(\frac{12}{2}\right)^{th} \text{ term}\right]$

$= \text{mean of } [(5)^{th} \text{ term and } (6)^{th} \text{ term}]$

$= \text{mean of } [24 \text{ and } 25]$

$= \frac{24 + 25}{2}$

$= \frac{49}{2}$

$= 24.5$

Hence, the median is 24.5

Solution 8: Arranging the given data in ascending order:

42, 43, 44, 44, $(2x + 3)$, 45, 45, 46, 47

Number of terms = 9 (odd)

$\therefore$ Median $= \left(\frac{n+1}{2}\right)$ th term

$\Rightarrow 45 = \left(\frac{9+1}{2}\right)$ th term

$\Rightarrow 45 = \left(\frac{10}{2}\right)$ th term

$\Rightarrow 45 = (5)$th term

$\Rightarrow 45 = 2x + 3$

$\Rightarrow 2x = 45 - 3$

$\Rightarrow 2x = 42$

$\Rightarrow x = 21$

Hence, the value of x is 21.

Arranging the given data in ascending order:

42, 43, 44, 44, 45, 45, 45, 46, 47

Here, 45 occurs maximum number of times.

$\therefore$ Mode $= 45$

Hence, the mode of the data is 45.

Multiple Choice Questions (MCQ)

Solution 1: (c) 7

Mean of 5 observations = 11

We know:

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$$

$$\Rightarrow 11 = \frac{x + x + 2 + x + 4 + x + 6 + x + 8}{5}$$

$\Rightarrow 11 = 5x + 205$

$\Rightarrow 5x + 20 = 55$

$\Rightarrow 5x = 35$

$\Rightarrow x = 7$

Solution 2: (c) $11\frac{1}{3}$

Mean of 5 observations = 9

We know:

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$$

$$\Rightarrow 9 = \frac{x + x + 3 + x + 5 + x + 7 + x + 10}{5}$$

$\Rightarrow 9 = 5x + 255$

$\Rightarrow 5x + 25 = 45$

$\Rightarrow 5x = 20$

$\Rightarrow x = 4$

Therefore, the last three observations are $(4 + 5)$, $(4 + 7)$ and $(4 + 10)$, i.e.,

9, 11 and 14.

Now,

$$\text{Mean of the last three terms} = \frac{9 + 11 + 14}{3} = \frac{34}{3} = 11\frac{1}{3}$$

Solution 3: (b) 0

If $\bar{x}$ is the mean of $x_1, x_2, x_3, \ldots, x_n$, then, we have

$$\sum_{i=1}^{n} x_i = \bar{x}$$

$$\Rightarrow \sum_{i=1}^{n} x_i - \bar{x} = 0$$

Solution 4: (b) is decreased by 8

Solution 5: (c) 53 kg

Mean weight of six boys = 48 kg

Let the weight of the 6th boy be x kg.

We know:

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Total number of observations}}$$

$$= \frac{51 + 45 + 49 + 46 + 44 + x}{6}$$

$$= \frac{235 + x}{6}$$

Given:

Mean = 48 kg

$$\Rightarrow \frac{235 + x}{6} = 48$$

$$\Rightarrow 235 + x = 288$$

$$\Rightarrow x = 53$$

Hence, the weight of the 6th boy is 53 kg.

Solution 6: (b) 39.4

Mean of the marks scored by 50 students = 39

Sum of the marks scored by 50 students $= (39 \times 50) = 1950$

Correct sum $= (1950 + 43 - 23) = 1970$

$\therefore$ Mean $= \frac{1970}{50} = 39.4$

Solution 7: (c) 64.91

Mean of 100 items = 64

Sum of 100 items $= 64 \times 100 = 6400$

Correct sum $= (6400 + 36 + 90 - 26 - 9) = 6491$

Correct mean $= \frac{6491}{100} = 64.91$

Solution 8: (b) 51

Mean of 100 observations = 50

Sum of 100 observations $= 100 \times 50 = 5000$

It is given that one of the observations, 50, is replaced by 150.

$\therefore$ New sum $= (5000 - 50 + 150) = 5100$

And,

Resulting mean $= \frac{5100}{100} = 51$

Solution 9: (b) $\frac{1}{2}(\bar{x} + \bar{y})$

$$\bar{z} = \frac{(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)}{2n}$$

Given,

$$\bar{x} = \frac{x_1, x_2, \dots, x_n}{n}$$

$\Rightarrow x_1, x_2, \dots, x_n = n\bar{x}$

$$\bar{y} = \frac{y_1, y_2, \dots, y_n}{n}$$

$$\Rightarrow y_1, y_2, \dots, y_n = n\bar{y}$$

$$\therefore \bar{z} = \frac{(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)}{2n}$$

$$= \frac{n\bar{x} + n\bar{y}}{2n}$$

$$= \frac{\bar{x} + \bar{y}}{2}$$

$$= \frac{1}{2}(\bar{x} + \bar{y})$$

Solution 10: (b) $\left(a + \frac{1}{a}\right)\frac{\bar{x}}{2}$

$$\text{Required mean} = \frac{(ax_1, ax_2, \dots, ax_n) + \left(\frac{x_1}{a}, \frac{x_2}{a}, \dots, \frac{x_n}{a}\right)}{2n}$$

$$= \frac{1}{2}\left\{\frac{a(x_1, x_2, \dots, x_n)}{n} + \frac{\frac{1}{a}\{(x_1, x_2, \dots x_n)\}}{n}\right\}$$

$$= \frac{1}{2}\left\{a\bar{x} + \frac{1}{a}\bar{x}\right\}$$

$$= \left(a + \frac{1}{a}\right)\frac{\bar{x}}{2}$$

Solution 11: (c) $\frac{\sum_{i=1}^{n} n_i \bar{x}_i}{\sum_{i=1}^{n} n_i}$

Sum of all terms $= n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_n\bar{x}_n$

Number of terms $(n_1 + n_2 + \dots + n_n)$

$$\therefore \text{Mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_n\bar{x}_n}{n_1 + n_2 + \dots + n_n} = \frac{\sum_{i=1}^{n} n_i \bar{x}_i}{\sum_{i=1}^{n} n_i}$$

Solution 12: (c) 25

x	y	x × y
3	6	18
5	8	40
7	15	105
9	p	$9p$
11	8	88
13	4	52
Total	$41 + p$	$303 + 9p$

Now,

$$\text{Mean} = \frac{303 + 9p}{41 + p}$$

Given:

Mean = 8

$$\therefore \frac{303 + 9p}{41 + p} = 8$$

$\Rightarrow 303 + 9p = 328 + 8p$

$\Rightarrow p = 25$

Solution 13: (b) 29

Arranging the weight of 10 students in ascending order, we have:

0, 13, 15, 20, 27, 29, 31, 34, 43, 50, 56

Here, n is 11, which is an odd number.

Thus, we have:

$$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation Median score}$$

$$= \text{Value of } \left(\frac{11+1}{2}\right)^{\text{th}} \text{ term}$$

$= \text{Value of } \left(\frac{12}{2}\right)^{\text{th}} \text{ term}$

$= \text{Value of } 6^{\text{th}} \text{ term}$

$= 29$

Solution 14: (c) 42 kg

Arranging the numbers in ascending order, we have:

31, 35, 36, 38, 40, 44, 45, 52, 55, 60

Here, n is 10, which is an even number.

Thus, we have:

$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)^{\text{th}} \text{ observation } \& \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observation}$

$\text{Median weight} = \text{Mean of the weights of } \left(\frac{10}{2}\right) \text{th student } \& \left(\frac{10}{2}+1\right)^{\text{th}} \text{ student}$

$= \text{Mean of the weights of } 5^{\text{th}} \text{ student } \& \ 6^{\text{th}} \text{ student}$

$= 12(40+44)$

$= 42$

Hence, the median weight is 42 kg.

Solution 15: (c) 6

We will arrange the given data in ascending order as:

3, 4, 4, 5, 6, 7, 7, 7, 12

Here, n is 9, which is an odd number.

Thus, we have:

$\text{Median} = \text{Value of } \frac{1}{2}(n+1)^{\text{th}} \text{ term}$

$\text{Median score} = \frac{1}{2}(9+1)\text{th term} = 5^{\text{th}} \text{ term} = 6$

Solution 16: (c) 54

We will arrange the data in ascending order as:

22, 34, 39, 45, 54, 54, 56, 68, 78, 84

Here, n is 10, which is an even number.

Thus, we have:

$$\text{Median} = \text{Mean of } \left(\frac{n}{2}\right)\text{th } \& \left(\frac{n}{2}+1\right)\text{th observations}$$

$$= 12\left(5^{\text{th}} \text{ observation} + 6^{\text{th}} \text{ observation}\right)$$

$$= 12(54+54)$$

$$= 54$$

Solution 17: (b) 15

Here, 14 occurs 4 times, 15 occurs 5 times, 16 occurs 1 time, 18 occurs 1 time, 19 occurs 1 time and 20 occurs 1 time. Therefore, the mode, which is the most occurring item, is 15.

Solution 18: (b) 21

The given data is in ascending order.

Here, n is 10, which is an even number.

Thus, we have:

$$\text{Median } = \text{Mean of } (n^2)^{\text{th}} \ \& \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observations}$$

$$= \frac{1}{2}(5^{\text{th}} \text{ observation} + 6^{\text{th}} \text{ observation})$$

$$= \frac{1}{2}(x+2+x+4)$$

$$= (x+3)$$

$$= 24$$

Also,

$x + 3 = 24$

$\Rightarrow x = 21$